

Hybrid 10/100 Gbps laser inter-satellite link planning in optical satellite networks

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With the continuously decreasing cost of launching satellites, low-Earth-orbit (LEO) optical satellite networks (OSNs) have emerged as an important research topic. By using laser communication, 10 Gbps laser inter-satellite links (LISLs) have been fully deployed, while 100 Gbps LISLs are rapidly advancing and are expected to become key components of future networks. With the expected growth in satellite-delivered service demand, OSNs will inevitably enter a hybrid stage in which 10 and 100 Gbps LISLs coexist. However, such coexistence may cause traffic bottlenecks of 10 Gbps LISLs and underutilization of 100 Gbps LISLs in LEO OSNs. From a network planning perspective, this paper focuses on developing efficient hybrid 10/100 Gbps LISL planning algorithms. We first propose a uniformly connected 100 Gbps LISL planning algorithm (UC-100GPA) to ensure the widespread deployment of 100 Gbps LISLs across the network. Based on this, a genetic algorithm for 100 Gbps LISL planning optimization (GA-100GPO) is proposed to further determine the sub-optimal deployment ratio of 100 Gbps LISLs in both the intra-orbit plane (intra-OP) and the inter-orbit plane (inter-OP). Simulation results show that UC-100GPA reduces the blocking ratio by 30.32% and 14.88% compared to deployments without 100 Gbps LISLs and with intra-OP 100 Gbps LISLs, respectively. When the traffic load is 800 Erlang, GA-100GPO achieves a blocking ratio of only 0.56% under a sub-optimal deployment ratio of 63.38% intra-OP and 36.62% inter-OP. © 2026 Optica Publishing Group. All rights, including for text and data mining (TDM), Artificial Intelligence (AI) training, and similar technologies, are reserved.

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1. INTRODUCTION

With the growing demand for global communications and rapid development of information technology, optical satellite networks (OSNs) are gradually becoming an important infrastructure for future communications [1]. OSNs typically consist of large satellite constellations operating in low Earth orbit (LEO). These constellations provide high-throughput, low-latency services to regions lacking terrestrial infrastructure, such as remote rural areas, oceans, and deserts [2]. Consequently, OSNs play an important role in extending global Internet coverage and narrowing the digital divide [3]. Motivated by its importance, several major initiatives have accelerated the deployment of LEO systems. For instance, Starlink aims to deploy about 42,000 satellites with advanced laser communication. As of December 2025, Starlink has launched over 10,000 satellites, of which more than 9400 are successfully in orbit [4]. These satellites provide a global Internet coverage in more than 100 countries and regions, with a user base exceeding nine million. Rapid development

highlights that LEO OSNs are gradually becoming a critical infrastructure for future networks.

In recent years, advances in laser communication technology have drawn increasing attention to laser inter-satellite links (LISLs). These links offer several advantages, including high data rates, strong resistance to interference, high reliability, and enhanced security [5]. These advantages make OSNs a promising solution for future LEO OSNs and a key focus of both academia and industry. This potential is already reflected in practice, as 10 Gbps LISLs have been widely deployed in OSNs, providing stable and cost-effective performance for user services [6,7]. Meanwhile, recent breakthroughs in space laser communication technology have accelerated the development of 100 Gbps LISLs, which are planned for future OSNs and are expected to become the mainstream [8]. On one hand, deploying full 10 Gbps LISLs across the OSNs fails to meet the growing demand for high-bandwidth services. On the other hand, deploying full 100 Gbps LISLs across the network results in high deployment costs and resource inefficiency. With the integration of higher-capacity LISLs into newly launched LEO

satellites, future OSNs are expected to enter a stage featuring the coexistence of 10 and 100 Gbps LISLs.

However, in such hybrid deployments, the limited capacity of 10 Gbps LISLs can be rapidly exhausted under high traffic demand, thereby leading to potential blocking bottlenecks. In contrast, the capacity of 100 Gbps LISLs may not be fully utilized, as typical traffic demand often falls short of their maximum throughput, resulting in inefficient resource utilization. Figure 1 shows an illustration of the hybrid deployment of 10/100 Gbps LISLs in OSNs. Solid lines denote intra-orbit plane (intra-OP) LISLs, while dashed lines denote inter-orbit plane (inter-OP) LISLs. Black and blue lines represent 10 and 100 Gbps LISLs, respectively, and red dashed lines indicate congested 10 Gbps LISLs. The difference between low-capacity and high-capacity LISLs makes traffic distribution unbalanced in OSNs. When traffic is concentrated on 10 Gbps LISLs, these LISLs experience severe capacity pressure, leading to local congestion while part of the 100 Gbps capacity remains underutilized. The lack of an adaptive planning mechanism further causes resource underutilization and increases the likelihood of traffic bottlenecks. Therefore, there is an urgent need to explore efficient hybrid 10/100 Gbps LISL planning algorithms to enhance the network performance of OSNs. Given that LISLs are scarce and valuable transmission resources in space, efficiently managing the coexistence of 10/100 Gbps LISLs in LEO OSNs is a critical problem.

From a network planning perspective, this paper investigates the problem of hybrid 10/100 Gbps LISL planning in OSNs and develops efficient algorithms to improve overall network performance. Simulation results show that the proposed algorithms in this paper can significantly improve the network performance of OSNs.

The contributions of our work include the following:

(1) We formulate and investigate a novel hybrid 10/100 Gbps

LISL planning problem in OSNs, where the coexistence of 10/100 Gbps LISLs leads to traffic bottlenecks of 10 Gbps LISLs and underutilization of 100 Gbps LISLs. (2) The dedicated 100 Gbps LISL planning algorithms are proposed. A uniformly connected 100 Gbps LISL planning algorithm (UC-100GPA) is developed to mitigate traffic bottlenecks by constructing a balanced 100 Gbps connectivity topology, and a genetic algorithm for 100 Gbps LISL planning optimization (GA-100GPO) is further proposed to determine a sub-optimal deployment ratio of 100 Gbps LISLs between the intra-OP and the inter-OP. (3) Extensive simulations are conducted under different traffic loads and deployment scenarios. Compared with baseline deployment schemes, the proposed algorithms achieve noticeable improvements in blocking ratio, average latency, and throughput, demonstrating the effectiveness of the proposed 100 Gbps LISL planning algorithm.

The rest of this paper is organized as follows. Section 2 introduces the related work of this paper. Section 3 introduces the OSN model used in this paper. Section 4 describes the 100 Gbps LISL planning algorithms proposed in this paper. Simulation results are demonstrated in Section 5. Finally, Section 6 concludes this paper.

2. RELATED WORK

A. Virtual Topology

Owing to the high mobility of satellites, the topology of the OSN changes rapidly over time. Virtual topology approaches [9] address this challenge by dividing dynamic network topology into a sequence of time slices. Each time slice represents a static topology that remains unchanged for a short interval. Based on this framework, [10] discretized the time-varying topology into slices and studied rerouting mechanisms under

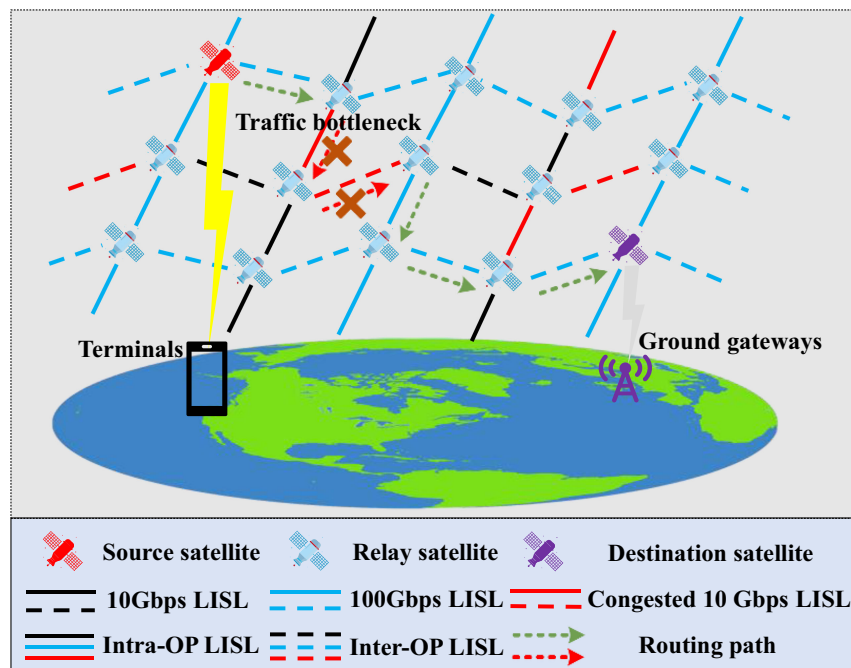


Fig. 1. Hybrid deployment of 10/100 Gbps LISLs in OSNs.

node failures, leveraging the predictable nature of OSN dynamics to precompute routing paths. Moreover, [11] investigated the trade-off between the accuracy of the slice and storage overhead. Based on these studies, we focus on the planning of hybrid 10/100 Gbps LISLs within a virtual-topology framework in this paper.

B. Static Link Planning and Topology Design

At present, numerous studies [12–16] have been conducted on the topology design and network planning in terrestrial networks. In satellite networks, most corresponding studies focus on optimizing LISL planning, which is supposed to have one type of unified transmission capacity of LISLs. Static link planning and topology design issues caused by the continuous movement of LEO satellites in satellite networks have been widely studied. For example, Guo *et al.* [17] proposed two algorithms, which aim to maximize throughput by reducing the average path length and improving the inter-satellite link (ISL) utilization. Additionally, Hu *et al.* [18] presented a two-stage topology design method for LEO mega-constellations, focusing on optimizing network performance by reducing latency and improving robustness. Moreover, Yang *et al.* [19] analyzed ISL design for mega-constellation networks, proposing a method for selecting ISLs based on laser terminal acquisition probability and using a nonlinear optimization model to improve link stability. In addition, Chaudhry and Yanikomeroglu [20] conducted a comprehensive classification and analysis of LISLs in a Starlink constellation. By categorizing LISLs according to orbital-plane relationships and link duration, they investigated the impact of LISL range on the number and types of feasible LISLs. Their study provides valuable insights into the characteristics of LISL-based topologies. Furthermore, Chaudhry *et al.* [21] studied the impact of LISL range on network connectivity and end-to-end latency in OSNs. By assuming a uniform LISL range for all satellites, they showed that increasing the link range enhances network connectivity, reduces hop count, and significantly lowers average latency for intercontinental communications.

C. Dynamic Link Planning and Resource Scheduling

Furthermore, dynamic link planning and resource scheduling issues in satellite networks are also intensively investigated. Our previous work explored efficient topology design in OSNs. We [22] proposed an ISL removal algorithm to remove parts of ISLs from the grid-mesh topology, thereby significantly enhancing average bandwidth utilization while maintaining reliability. In addition, we also introduced the concept of laser ON/OFF control for energy efficiency in OSNs [23]. By employing a gravity-based traffic abstraction (GNTA) model, we assessed the importance of ISLs and proposed a GNTA-based ON/OFF control algorithm that dynamically switches off underutilized ISLs to enhance energy efficiency. Similarly, Bhattacharjee and Singla [24] introduced a motif-based topology design framework for large-scale LEO constellations, aiming to balance link stability and latency under extremely high orbital dynamics. Besides, Li *et al.* [25] proposed a joint optimization algorithm, DeepISL, to conduct ISL and power

allocation planning in LEO satellite networks. While the approach performs well in dynamic scenarios, its effectiveness in large-scale real-world networks remains an area for future research. In addition, Leyva-Mayorga *et al.* [26] proposed a weighted dynamic matching and resource allocation framework to maximize the sum rate of inter-plane inter-satellite links in dense LEO constellations. What is more, Liu *et al.* [27] formulated the optical inter-satellite link assignment problem as a perfect matching model in graph theory and proposed the LAA-PMM algorithm to optimize link utilization and path length. Moreover, Chen *et al.* [28] proposed an ISL re-planning algorithm via multi-agent deep reinforcement learning for LEO satellite constellations to address link failures. Bhattacharjee *et al.* [29] investigated on-demand routing in LEO mega-constellations with dynamic laser inter-satellite links, explicitly considering LISL setup delay as a penalty in end-to-end latency. By formulating a routing optimization problem and proposing several heuristic algorithms, they demonstrated the trade-offs between latency performance and computational complexity under dynamic LISL establishment. Besides, they [30] investigated the feasibility of dynamic LISLs by quantifying LISL setup delay and incorporating it into the end-to-end latency analysis of OSNs. They quantified the latency degradation caused by frequent path changes and identified the maximum tolerable LISL setup delay for dynamic LISLs.

In summary, although extensive efforts have been devoted to ISL planning and optimization in satellite networks, existing studies predominantly assume homogeneous link capacities. The problem of hybrid 10/100 Gbps LISL planning in OSNs, particularly from a network planning perspective, remains unexplored. This work addresses this problem by proposing dedicated 100 Gbps LISL planning algorithms and demonstrating their effectiveness in mitigating traffic bottlenecks and improving network performance.

3. OSN MODEL

In this section, we introduce the model of OSNs, including the satellite constellation, orbital parameters, network topology, traffic model, and service model.

A. Satellite Constellation

The constellation configuration is a key aspect of OSN design. Satellites can operate in both circular and elliptical orbits. In this paper, a circular orbit is adopted for subsequent analysis because of its constant altitude [31]. Based on circular orbits in OSNs, satellites are supposed to be uniformly distributed in each orbital plane. A homogeneous constellation can be described by a four-tuple of parameters $\{alt, inc, N_{orb}, N_{sat}\}$, respectively representing the orbital altitude, the orbital inclination, the number of orbital planes, and the number of satellites in each orbit plane. In each orbital plane, the distance between neighboring satellites remains unchanged due to each satellite having the same speed. However, the distance between adjacent satellites in different orbits changes over time due to their complex spatial relationship. Additionally, the phase factor, which represents the relative position difference

Table 1. Notations Description

Notations	Description	Notations	Description
alt	Orbital altitude	$Erlang$	Traffic load
inc	Orbital inclination	μ	Service rate
N_{orb}	Number of orbital planes	L_{total}	Total number of 100 Gbps LISLs to be deployed
N_{sat}	Number of satellites per plane	L_{intra}	Number of intra-OP 100 Gbps LISLs to be deployed
G	Undirected graph	L_{inter}	Number of inter-OP 100 Gbps LISLs to be deployed
V	Set of satellite nodes	r_{intra}	Ratio of 100 Gbps LISLs deployed in the intra-OP
E	Set of LISLs	r_{inter}	Ratio of 100 Gbps LISLs deployed in the inter-OP
P_i	Population of the i city	f	Objective function
P_j	Population of the j city	p_c	Crossover probability
$D_{i,j}$	Distance between the i and j cities	p_m	Mutation probability
$H_{i,j}$	Traffic between the i and j cities	α	Weight factor of crossover
A	Gravitational constant factor	σ	Mutation strength
λ	Arrival rate	r^*	Sub-optimal intra-OP deployment ratio
f^*	Sub-optimal fitness value	N_p	Population size
Q	Set of population	K	Maximum number of generations
Q'	Set of updated population	r_i	The i individual in the set Q
$N_{blocked}$	The number of blocked service requests	$N_{success}$	The number of successfully transmitted services
N_{total}	The total number of service requests	B_i	The bandwidth of the i service request
T_i	Duration of the i service	T_{sim}	Total simulation time

between satellites in adjacent planes, influences the distances between satellites and is crucial for constructing a constellation with uniform coverage capability [32]. The above constellation parameters provide a basis for establishing LISLs, designing topology, and planning satellite networks. A list of notations in this paper is shown in Table 1.

B. Network Topology

In this paper, a network topology is built based on the above satellite constellations. Considering the predictable characteristics of satellite constellations, we represent the topology of OSNs as an undirected graph $G(V, E)$ [33]. V is the set of satellites, and each satellite v is identified with two indexes—the orbit-id i and the satellite-id j inside each orbital plane, i.e., $\{v_{i,j} \in V \mid 0 \leq i < N_{orb}, 0 \leq j < N_{sat}\}$. E is the set of LISLs, including intra-OP LISLs and inter-OP LISLs. Each LISL has its specific length and bandwidth. Each satellite is assumed to be equipped with four laser terminals [34]. Accordingly, it can communicate with four neighboring satellites: two adjacent satellites in the same orbital plane and two in neighboring orbital planes. Therefore, each satellite has two intra-OP LISLs and two inter-OP LISLs. Intra-OP LISLs can be further identified by two indices, orbit-id i and satellite-id j in each orbital plane, i.e., $\{l_{i,j}^a \in E_a \mid 0 \leq i < N_{orb}, 0 \leq j < N_{sat}\}$. $l_{i,j}^a$ is the intra-OP LISLs between the j satellite and the $j+1$ satellite in the i orbital plane. Similarly, inter-OP LISLs can be further identified by two indices, orbit-id i and satellite-id j in each orbital plane, i.e., $\{l_{i,j}^r \in E_a \mid 0 \leq i < N_{orb}, 0 \leq j < N_{sat}\}$. $l_{i,j}^r$ is the inter-OP LISLs between the j satellite in the i orbital plane and the j satellite in the $i+1$ orbital plane. For each intra-OP LISL and inter-OP LISL, we define its bandwidth capacities as $C_{i,j}^a$ and $C_{i,j}^r$, respectively. Furthermore, each undirected graph can be divided into several time slices. Within each time slice, the spatial positions of satellite nodes and LISLs are assumed to remain unchanged. Under this assumption, the

dynamic OSN topology can be represented as a sequence of static topologies. For each time slice, the routing algorithm computes optimal transmission paths for service requests and allocates the corresponding bandwidth resources.

C. Traffic Model and Service Model

To evaluate the performance of OSNs, this paper constructs a traffic model to simulate user traffic as input to the network. Considering the critical role of OSNs in future global data transmission, this paper focuses on scenarios in which all traffic is transmitted through LISLs, access links, and feeder links.

In OSNs, traffic is supposed to come from the ground. However, traffic is distributed unevenly in different areas around the world. Thus, it is necessary to prioritize the population in traffic allocation. Larger populations in developed cities demand more traffic, while smaller populations in underdeveloped cities require less. Therefore, traffic distribution is partly defined by population density. In addition, traffic is also affected by the distance. When the distance between two cities is short, they tend to generate more traffic. Based on this principle, this paper uses a gravity-based traffic model to determine the traffic demand between city pairs.

The gravity-based traffic model is derived from the world trade theory and provides traffic between city pairs, accounting for population and distance factors. It is shown in Eq. (1) [35]:

$$H_{i,j} = A \frac{P_i \cdot P_j}{D_{i,j}}, \quad (1)$$

where P_i and P_j denote the population of the i city and j city, respectively. $D_{i,j}$ denotes the geographic distance between two cities and $H_{i,j}$ denotes the traffic between city pairs. Besides, the gravitational constant factor A is a normalization parameter in the gravity-based traffic formulation. For simplicity, A is set to 1 in this paper since its value does not affect the results of traffic calculation. It is worth noting that this is a very general model that only considers population and distance.

In reality, traffic is affected by many factors, including a city's level of economic development, Internet access rates across different countries, and so on. In order not to lose the generality, we follow the traditional model to estimate traffic between city pairs.

Moreover, the arrival rate and service duration are simulated using the Irish model, where the arrival rate of service obeys a Poisson distribution parameterized as λ and the service duration obeys a negative exponential distribution parameterized as μ [36]. Traffic load *Erlang* is jointly determined by arrival rate and service duration.

4. 100 Gbps LISL PLANNING ALGORITHM

To minimize traffic bottlenecks caused by hybrid 10/100 Gbps LISL planning in OSNs, we first propose the UC-100GPA algorithm to improve the network performance of OSNs. On this basis, we propose the GA-100GPO algorithm to further determine the deployment ratio of 100 Gbps LISLs in the intra-OP and inter-OP.

A. Uniformly Connected 100 Gbps LISL Planning Algorithm

In OSNs, the efficient transmission of services relies on both intra-OP and inter-OP LISLs. Deployment algorithms that rely exclusively on either intra-OP or inter-OP 100 Gbps LISLs may result in significant performance degradation because of insufficient network connectivity. Thus, an optimized algorithm for deploying 100 Gbps LISLs in both the intra-OP and inter-OP is essential to ensure optimal network performance and minimize traffic bottlenecks.

To efficiently deploy 100 Gbps LISLs in OSNs, this paper proposes a uniformly connected 100 Gbps LISL planning algorithm (UC-100GPA) to address traffic bottlenecks caused by 10 Gbps LISLs. The UC-100GPA algorithm adopts uniformity and connectivity as its two core principles, enabling balanced and efficient deployment of 100 Gbps LISLs.

The uniformity principle ensures that 100 Gbps LISLs are distributed as evenly as possible across the satellite constellation. This principle maximizes the coverage of 100 Gbps LISLs, mitigates localized performance degradation, and ensures service availability across most areas of OSNs. The connectivity principle focuses on optimizing the ratio of intra-OP and inter-OP 100 Gbps LISLs. This ratio is critical for ensuring that both vertical and horizontal dimensions of the network are adequately equipped with 100 Gbps LISLs. By maintaining a balance between intra-OP and inter-OP deployments, the UC-100GPA algorithm ensures the formation of a connected 100 Gbps LISL topology across the satellite constellation. This facilitates service transmission between different orbit planes and reduces the risk of traffic bottlenecks.

Figure 2 illustrates an example deployment generated by the UC-100GPA algorithm, which visually demonstrates how uniformly connected 100 Gbps LISLs help mitigate traffic bottlenecks caused by hybrid 10/100 Gbps LISL planning. In Fig. 2, green lines denote 100 Gbps LISLs deployed by the UC-100GPA algorithm, black lines denote 10 Gbps LISLs, and red lines highlight congested 10 Gbps LISLs under hybrid

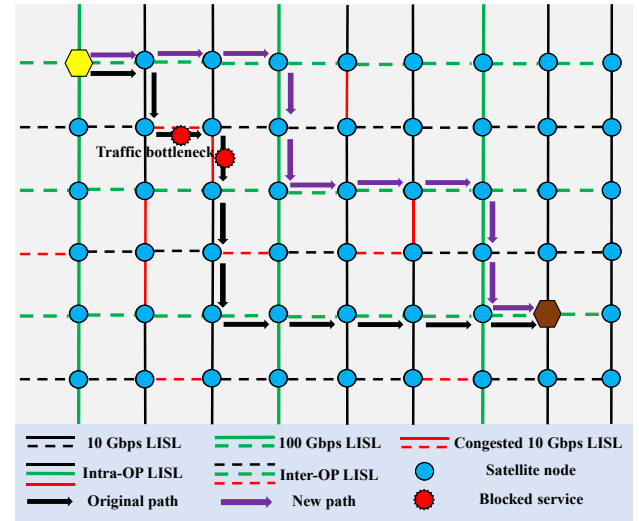


Fig. 2. Topology of the UC-100GPA algorithm.

deployment. This figure serves as an illustrative example to facilitate understanding of the UC-100GPA algorithm. Solid lines denote intra-OP LISLs, while dashed lines denote inter-OP LISLs. The original routing path and new path are marked by black and purple arrows, respectively. As shown in Fig. 2, the deployment of 100 Gbps LISLs is important for the successful delivery of services. When 100 Gbps LISLs are unevenly distributed or disconnected, a large number of 10 Gbps LISLs will concentrate in certain areas, causing traffic bottlenecks and resulting in service blockage, which can be seen in the original path. By considering principles of uniformity and connectivity, the UC-100GPA algorithm can achieve a uniformly connected distribution of 100 Gbps LISLs in both the intra-OP and inter-OP. In this way, the algorithm can reduce congestion, maximize the utilization of 100 Gbps LISLs, and improve overall network performance, as shown in the new path in the figure.

The detailed procedures of the UC-100GPA algorithm are summarized in Algorithm 1.

The UC-100GPA algorithm operates as follows. First, the total number of 100 Gbps LISLs to be deployed L_{total} and the deployment ratio of 100 Gbps LISLs in the intra-OP r_{intra} are determined, which can be used to calculate the number of 100 Gbps LISLs deployed in the intra-OP L_{intra} and inter-OP L_{inter} (Steps 2–3). Then, a subset of orbital planes is first selected at regular intervals (i.e., the 10th, 20th, and 30th planes) according to L_{intra} . All intra-OP LISLs in these selected orbital planes are subsequently upgraded to 100 Gbps (Steps 4–6). Similarly, a subset of satellite indices within each orbital plane is selected at regular intervals (i.e., the 5th, 10th, 15th, and 20th satellites) according to L_{inter} . All inter-OP LISLs of the selected satellite indices in each orbital plane are then upgraded to 100 Gbps, ensuring interconnections with the upgraded intra-OP LISLs (Steps 7–9). Therefore, a connected graph of 100 Gbps LISLs is formed, as is shown by the green links in Fig. 2. If any LISL is disconnected from other LISLs, the algorithm will adjust its position until it is connected with others (Steps 10–14). Finally, the algorithm will output a final connectivity graph (Step 15). The UC-100GPA

Algorithm 1. UC-100GPA Algorithm

Input: L_{total} - Total number of 100 Gbps LISLs to be deployed
 r_{intra} - Deployment ratio of 100 Gbps LISL in intra-OP
Output: Connectivity graph of the constellation topology with optimized deployment of 100 Gbps LISLs

```

1: Func deployLISLs ( $L_{\text{total}}, r_{\text{intra}}$ )
2:    $L_{\text{intra}} \leftarrow L_{\text{total}} * r_{\text{intra}}$ 
3:    $L_{\text{inter}} \leftarrow L_{\text{total}} - L_{\text{intra}}$ 
4:   foreach orbitalPlane in intra-OP do
5:     distribute  $L_{\text{intra}}$  uniformly across orbitalPlane
6:   end
7:   foreach orbitalPlane in inter-OP do
8:     place  $L_{\text{inter}}$  uniformly to connect with intra-OP LISLs
9:   end
10:  foreach LISL in allLinks do
11:    if LISL is not connected to any intra-OP or inter-OP LISL then
12:      adjust position of laserLink
13:    end
14:  end
15:  return final connectivity graph
16: end

```

algorithm adheres to both uniformity and connectivity principles, ensuring optimal coverage, high LISL utilization, and reduced traffic bottlenecks caused by 10 Gbps LISLs.

In this paper, the UC-100GPA algorithm is compared with four baseline deployment schemes under the same total number of deployed 100 Gbps LISLs. This comparison aims to isolate the impact of different deployment schemes, excluding the effect of the number of deployed 100 Gbps LISLs. Four topology schemes to be compared with the UC-100GPA algorithm are shown in Fig. 3.

It is worth noting that the four schemes are not directly adopted from prior work, but are deliberately designed as baseline and extreme cases to illustrate the impact of different deployment schemes. Specifically, scheme (a) represents intra-OP deployment, where 100 Gbps LISLs are deployed only in the intra-OP, while scheme (b) represents inter-OP

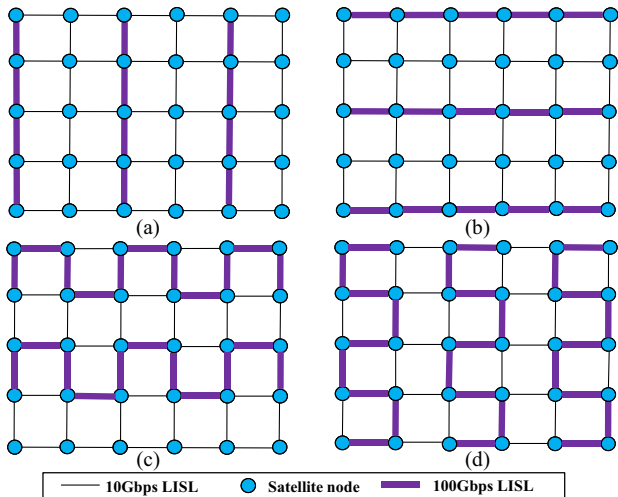


Fig. 3. Diagram of comparison topology schemes. (a) Intra-OP deployment, (b) inter-OP deployment, (c) uniform horizontal deployment, and (d) uniform vertical deployment.

deployment, where 100 Gbps LISLs are deployed only in the inter-OP. These two schemes correspond to single-type orbit extreme deployment cases. Schemes (c) and (d) represent uniform horizontal deployment and uniform vertical deployment, respectively, where 100 Gbps LISLs are uniformly deployed along one dimension of the grid topology, covering both the intra-OP and inter-OP. These two schemes serve as representative uniform deployment baselines.

Together, the four deployment schemes form a set of baseline and extreme deployment schemes that enable intuitive comparisons with the proposed UC-100GPA algorithm and help reveal the network performance of different 10/100 Gbps LISL planning schemes. In Section 5, we will simulate network performance for the UC-100GPA algorithm and the four schemes above.

For the schemes in Figs. 3(a) and 3(b), 100 Gbps LISLs are deployed exclusively in either the intra-OP or inter-OP, representing single-orbit deployment schemes. The schemes in Figs. 3(c) and 3(d) adopt mixed deployment schemes, where LISLs are distributed in both the intra-OP and inter-OP along one dimension of the grid topology. As a result, these schemes inherently have fixed intra-OP deployment ratios of 100 Gbps LISLs. However, the UC-100GPA algorithm considers a hybrid deployment of intra-OP and inter-OP 100 Gbps LISLs. To maintain generality, the intra-OP deployment ratio of 100 Gbps LISLs in the UC-100GPA algorithm is set to 0.5, representing a balanced deployment between intra-OP and inter-OP LISLs. In this case, under the same number of 100 Gbps LISLs, the differences in network performance reflect the effectiveness of the deployment scheme.

B. Genetic Algorithm for 100 Gbps LISL Planning Optimization

After determining the deployment of 100 Gbps LISLs using the UC-100GPA algorithm, it is necessary to determine the sub-optimal deployment ratio of 100 Gbps LISLs in the intra-OP and inter-OP to optimize network performance. In this subsection, we use the blocking ratio to reflect network performance. First, the objective function can be expressed using a mathematical formula, as illustrated in Eq. (2):

$$\text{Minimize } f(r_{\text{intra}}) = \text{Blocking Ratio}(r_{\text{intra}}, 1 - r_{\text{intra}}). \quad (2)$$

The problem is subject to the following statements:

$$0 \leq r_{\text{intra}} \leq 1, \quad (3)$$

$$r_{\text{intra}} + r_{\text{inter}} = 1. \quad (4)$$

Equation (3) means that the ratio of 100 Gbps LISLs deployed in the intra-OP r_{intra} ranges from 0 to 1. Eq. (4) indicates that the sum ratio of 100 Gbps LISLs deployed in the intra-OP and inter-OP should be equal to 1. Once r_{intra} is determined, r_{inter} is clear according to Eq. (4). The goal of this optimization problem is to search for a sub-optimal value of r_{intra} to minimize the blocking ratio in OSNs.

As shown in Eq. (2), r_{inter} is implicitly represented as $1 - r_{\text{intra}}$, ensuring that the blocking ratio accounts for both intra-OP and inter-OP 100 Gbps LISL deployments.

It is worth noting that BlockingRatio is not an analytical expression, but rather a performance metric obtained from network-level simulations. For a given value of r_{intra} , constellation topology is first constructed according to the proposed algorithms, with a deployment ratio of 100 Gbps LISL for the intra-OP and inter-OP. Service requests are then generated and routed based on the current network snapshot. The blocking ratio is the number of blocked service requests divided by the total number of service requests during the simulation period. This simulation-based definition is consistent with the formal blocking ratio formulation in Eq. (8).

The computational complexity of the problem is analyzed by considering a decision version with fixed traffic demand and 100 Gbps LISLs. The objective is to determine whether the blocking ratio can be reduced below a given threshold. This decision problem is equivalent to a capacitated network design problem (CNDP) with fixed-capacity links, in which link selection determines the available network capacity and the feasibility of routing under capacity constraints. Since CNDP is NP-hard [37], and this decision version can be reduced to the proposed LISL planning problem in polynomial time, the 100 Gbps LISL planning problem is NP-hard. Moreover, the use of simulation-based, non-analytic objective functions further increases the computational complexity. Therefore, we propose a genetic algorithm for 100 Gbps LISL planning optimization (GA-100GPO) to search for a sub-optimal deployment ratio. The GA-100GPO algorithm can efficiently explore sub-optimal solutions of the NP-hard problem without relying on the gradient information, which is especially suitable for dealing with the 100 Gbps LISL planning problem in this paper. In this case, we can search for the sub-optimal deployment ratio of the 100 Gbps LISL in both the intra-OP and inter-OP to improve network performance. The detailed procedures of the GA-100GPO algorithm are summarized in Algorithm 2.

Specific steps of the GA-100GPO algorithm are listed as follows. The GA-100GPO algorithm starts by generating an initial population of random values uniformly distributed between 0 and 1 (Step 1). Each value represents the ratio of 100 Gbps LISLs deployed in the intra-OP. The population size and the number of generations are then specified according to practical requirements. In each generation, the fitness value of each individual is evaluated based on the OSN's overall blocking ratio (Step 2), meaning that a lower blocking ratio can contribute to better network performance. After evaluating the current generation, the deployment ratio of the intra-OP and fitness value with the lowest blocking ratio are recorded (Step 3). Then, the GA-100GPO algorithm proceeds to the next iteration.

Before evaluating the fitness of the next generation, we need to generate new populations. Specifically, the top three individuals in the previous generation with the lowest blocking ratio are selected to retain the best solutions (Step 6). Then, three randomly generated individuals between 0 and 1 are added to increase population diversity (Step 7). Moreover, crossover and mutation are used to enhance population diversity, helping the algorithm avoid local optima and identify effective 100 Gbps LISL deployment ratios between the intra-OP and the inter-OP.

Algorithm 2. GA-100GPO Algorithm

Input: Population size N_p , maximum generations K , crossover probability p_c , mutation probability p_m
Output: Sub-optimal intra-OP deployment ratio r^* , corresponding fitness f^*

- 1: Initialize population $Q = \{r_1, r_2, \dots, r_n\}$, where each $r_i \in [0, 1]$ represents 100 Gbps LISL deployment ratio in intra-OP
- 2: Evaluate fitness $f(r_i) = \text{Blocking Ratio}(r_i, 1 - r_i)$ for all $r_i \in Q$
- 3: Record current best solution $(r^*, f^*) \leftarrow \text{argmin}(f_i)$
- 4: **for** generation = 1 to K **do**
- 5: Initialize new population $Q' \leftarrow \emptyset$
- 6: Add top 3 individuals with lowest fitness from Q into Q'
- 7: Add 3 randomly generated individuals $r_{\text{rand}} \in [0, 1]$ into Q'
- 8: **while** $|Q'| < N_p$ **do**
- 9: **if** $\text{rand}() < p_c$ **then**
- 10: Select two parents r_a, r_b from Q via roulette wheel selection
- 11: Generate child $r_c \leftarrow (r_a + r_b)/2$
- 12: **else**
- 13: Randomly select $r_c \in Q$
- 14: **if** $\text{rand}() < p_m$ **then**
- 15: Apply mutation: $r_c \leftarrow r_c + \text{Gaussian Noise}(0, \sigma)$
- 16: Clip r_c into $[0, 1]$ if necessary
- 17: Add r_c to Q'
- 18: Evaluate fitness $f_i = \text{BlockingRatio}(r_i)$ **for** all $r_i \in Q'$
- 19: Update $(r^*, f^*) \leftarrow \text{argmin}(f_i)$ **if** blocking ratio is lower than previous generation
- 20: $Q \leftarrow Q'$
- 21: **return** r^*, f^*

The remaining individuals in the new population are determined by crossover and mutation (Steps 8–17). In each generation, the remaining individuals are generated based on the two selected intra-OP deployment ratios of 100 Gbps LISLs from the previous generation. First of all, crossover is applied with a certain probability. In this algorithm, the selection probability of each individual is determined by its fitness value. Fitness values that result in lower blocking ratios in OSNs are more likely to be selected. Each individual's probability of being selected is illustrated in Eq. (5):

$$P(i) = \frac{1/f(r_i)}{\sum_{j=1}^N 1/f(r_j)}. \quad (5)$$

After two parent deployment ratios are chosen, a new deployment ratio can be calculated by crossover between the two parent deployment ratios. As is shown in Eq. (6), α denotes a random weight factor, typically assigned with a value of 0.5:

$$r_c = \alpha \cdot r_a + (1 - \alpha) \cdot r_b. \quad (6)$$

Therefore, the average of the two parent deployment ratios is used to generate a new individual. Furthermore, mutation is applied with a specified probability. Mutation introduces small variations (i.e., Gaussian noise) in the deployment ratio to explore a broader solution space, as illustrated in Eq. (7), where σ is the mutation strength:

$$r' = r + \mathcal{N}(0, \sigma). \quad (7)$$

In contrast, if no crossover or mutation occurs, a random deployment ratio from the previous generation is selected. This process continues until individuals reach the population's maximum. Fitness values, based on the blocking ratio of OSNs, are then computed for all individuals (Step 18). The GA-100GPO algorithm iterates over multiple generations, updating the sub-optimal deployment ratio of 100 Gbps LISLs in each generation (Step 19). At last, the GA-100GPO algorithm converges on the sub-optimal intra-OP deployment ratio that minimizes the blocking ratio and maximizes overall network performance. Finally, sub-optimal deployment ratios for the intra-OP and inter-OP in OSNs have been found through the GA-100GPO algorithm.

In each generation, the GA-100GPO algorithm evaluates the fitness of all individuals, with each individual's update process including elitism, random initialization, crossover, mutation, and so on. This process results in a per-generation complexity of $O(N_p)$. In addition, the algorithm undergoes K iterations, which results in complexity of $O(K)$. Consequently, the total time complexity of the GA-100GPO algorithm is $O(N_p * K)$.

In conclusion, the proposed UC-100GPA and GA-100GPO algorithms address the hybrid 10/100 Gbps LISL planning problem from two different but complementary perspectives. On one hand, UC-100GPA focuses on topology construction, which uniformly deploys 100 Gbps LISLs across the constellation and constructs a connectivity graph of 100 Gbps LISLs. Its objective is to mitigate traffic bottlenecks caused by 10 Gbps LISLs and to improve the overall connectivity of the network topology. On the other hand, GA-100GPO focuses on optimization of the deployment ratio, which searches for a sub-optimal deployment ratio of 100 Gbps LISLs between the intra-OP and the inter-OP that minimizes the blocking ratio and further improves network performance.

In practice, the sub-optimal deployment ratio obtained by GA-100GPO is used as input to UC-100GPA, which then generates the corresponding constellation topology. Together, the UC-100GPA and GA-100GPO algorithms form a two-stage planning framework for the hybrid 10/100 Gbps LISL planning problem.

5. SIMULATIONS AND ILLUSTRATIVE NUMERICAL RESULTS

To validate the effectiveness of the proposed algorithms, we conduct simulations using System Tool Kit (STK) and Python. STK is employed to model the satellite constellation and to provide time-varying satellite positions, inter-satellite distances, and dynamic link visibility information. Based on the geometric information provided by STK, a time-varying satellite network topology including satellites, ground gateways, LISLs, and satellite-ground links is generated [38]. Moreover, Python is used to implement a network-level simulation framework that generates service requests, executes routing algorithms, and collects network performance metrics.

Table 2. Simulation Parameters

Parameters	Values	Units
Constellation type	Walker-Delta	/
Phase factor	11	/
LISL bandwidth	10/100	Gbps
Bandwidth of service request	2500	Mbps
Number of ground gateway	10	/
Traffic load	200–1200	Erlang
Number of service requests	5000	/
Number of individuals	15	/
Number of iterations	30	/
Probability of crossover	80	%
Probability of mutation	40–80	%

A. Simulation Parameter Setting

In this paper, we built an OSN topology based on a Starlink constellation {550 km, 53°, 72, 22}. The constellation has an orbital altitude of 550 km, an orbital inclination of 53°, 72 orbital planes, and 22 satellites in each orbital plane. Accordingly, there are 1584 satellite nodes and 3168 LISLs in total. Each satellite node is supposed to be equipped with four laser terminals to connect its four-directional neighboring satellite nodes, thus forming two intra-OP LISLs and two inter-OP LISLs. The bandwidth of satellite-ground links is set to infinity in the simulation. In this way, LISLs are the only traffic bottleneck of network performance. The total simulation duration is set to 94 minutes, corresponding to one orbital period at the given orbital altitude, so that a complete and representative orbital cycle of the satellite network is captured. This setting ensures that the obtained performance metrics are representative. Table 2 lists the main parameters of the simulation in this paper.

The phase factor determines the relative phase offset between adjacent orbital planes and affects the spatial uniformity and symmetry of a Walker constellation. Although SpaceX has not publicly disclosed the exact phase factor used in Starlink, prior studies [39] have shown that a phase factor of 11 provides favorable coverage and geometric symmetry. Therefore, the phase factor is set to 11 in this work to ensure a well-balanced constellation topology.

Each service request is assigned a bandwidth of 2500 Mbps, which is obtained by aggregating 100 micro-flows of 25 Mbps into a single request, thereby reducing the total number of service requests and improving simulation efficiency. The number of ground gateways is set to 10 so that service requests are encouraged to traverse a large portion of the satellite network before downlinking, which helps emphasize the impact of hybrid 10/100 Gbps LISL deployment on network performance. The traffic load is varied from 200 to 1200 Erlang through trial simulations to represent moderate traffic loads, under which performance differences among deployment schemes can be clearly observed. In addition, the total number of service requests is set to 5000, a value under which the network operates in a saturated state and stable performance differences among deployment schemes can be reliably evaluated.

In the genetic algorithm adopted in this paper, the population evolves over 30 generations, with 15 individuals in each generation representing candidate intra-OP deployment ratios. These values are determined based on extensive trial simulations, which show that the algorithm consistently converges to a stable sub-optimal solution under this configuration. Further increasing the number of generations or population size does not lead to noticeable improvement in the final result, but incurs higher computational overhead. The crossover probability is set to 80% to encourage effective exchange of genetic information among individuals, while the mutation probability is designed to decrease gradually from 80% to 40% as evolution proceeds. Mutation is implemented by adding Gaussian noise with a standard deviation that decreases from 0.2 to 0.02. This adaptive strategy aims to maintain sufficient population diversity in the early stage, thereby enhancing the exploration capability and avoiding premature convergence to local optima. In the later stage, reduced mutation strength helps stabilize high-quality solutions and promotes smooth convergence toward a reliable sub-optimal solution.

To guarantee a global distribution of traffic in the simulation, we randomly select satellites as the source nodes of service requests. The destination nodes in the simulation are ground gateways located in 10 major cities worldwide. Based on the gravity-based traffic model introduced in Subsection 3.C, the service ratio of each city is determined by its relative traffic demand, which is derived from population and geographical distance. Each service request originates at a source satellite node and reaches the destination ground gateways via multiple LISLs and feeder links. The number of service requests is set to 5000. Moreover, the number of 100 Gbps LISLs to be deployed in OSNs is 800, which accounts for 25% of all LISLs in OSNs. This value is selected to represent a moderate deployment ratio of 100 Gbps LISLs. If the number of 100 Gbps LISLs is too small, the performance differences among different deployment schemes become insignificant. Conversely, if too many 100 Gbps LISLs are deployed, the performance of different schemes tends to converge, making it difficult to highlight the effectiveness of the proposed UC-100GPA algorithm. Through extensive trial simulations, 800 is selected to compare the network performance of different deployment schemes.

In this paper, the Dijkstra algorithm is used to compute the shortest routing path for each service request based on the current snapshot of the OSN. If no feasible path is available, the service request is regarded as blocked in the network. Otherwise, the corresponding transmission path is recorded, and bandwidth is allocated accordingly.

To evaluate the network performance, we consider three key performance metrics, the blocking ratio (BR), average latency (AL), and throughput (TP), in this paper. The BR is defined as the ratio of the number of blocked services to the total number of service requests, as is shown in Eq. (8):

$$BR = \frac{N_{\text{blocked}}}{N_{\text{total}}}. \quad (8)$$

Here, N_{blocked} denotes the number of blocked service requests and N_{total} denotes the total number of service requests.

Each service's latency is calculated as its routing path length divided by the speed of light, as shown in Eq. (9):

$$\text{Latency} = \frac{d_{\text{path}}}{c}. \quad (9)$$

The AL is defined as the average end-to-end transmission latency of all successfully transmitted services, as shown in Eq. (10):

$$AL = \frac{1}{N_{\text{success}}} \cdot \sum_{i=1}^{N_{\text{success}}} \frac{d_{\text{path}}}{c}, \quad (10)$$

where N_{success} is obtained by subtracting N_{blocked} from N_{total} , d_{path} denotes the path length of each successfully transmitted service, and c is the speed of light. Path length refers to the end-to-end propagation distance of the path, which is calculated as the sum of the physical distances of all LISLs and satellite-ground links along the routing path of the service request.

The TP is defined as the ratio of the total transmitted data volume of all successfully transmitted services to the total simulation time, where the transmitted data volume of each service is given by the product of its allocated bandwidth and service duration. It is shown in Eq. (11):

$$TP = \frac{\sum_{i=1}^{N_{\text{success}}} B_i T_i}{T_{\text{sim}}}, \quad (11)$$

where B_i denotes the bandwidth occupied by successfully transmitted services, T_i denotes the duration of service requests, and T_{sim} denotes total simulation time.

B. Results for Different Deployment Schemes

To evaluate the impact of the UC-100GPA algorithm on network performance, the traffic load is varied from 200 to 1200 Erlang with a step size of 200 Erlang. In this paper, the mean service duration is set to 200 s, corresponding to a session duration on the order of a few minutes (i.e., typical telephone call durations), which is commonly assumed in teletraffic modeling [40]. Similar assumptions on mean service duration have been widely adopted in traffic modeling studies and are sufficient for comparative performance evaluation. We set L_{total} as 800 and r_{intra} as 0.5 in the UC-100GPA algorithm. In this section, the deployment ratio of 100 Gbps LISLs for four deployment schemes and the UC-100GPA algorithm is 25%, which means that 100 Gbps LISLs account for 25% of all LISLs throughout the constellation.

As shown in Fig. 4, the blocking ratios under different 100 Gbps LISL deployment schemes all show significant reductions compared to no 100 Gbps LISL deployment. As the traffic load increases, the blocking ratio of different deployment schemes increases to some extent. However, the UC-100GPA algorithm has the lowest blocking ratio at all traffic loads. When the traffic load is 1000 Erlang, the blocking ratio of the UC-100GPA algorithm is only 1.7%, which is 30.32% lower than the scenario without 100 Gbps LISLs, and 14.88% lower than that of intra-OP deployment. The UC-100GPA algorithm achieves balanced deployment of

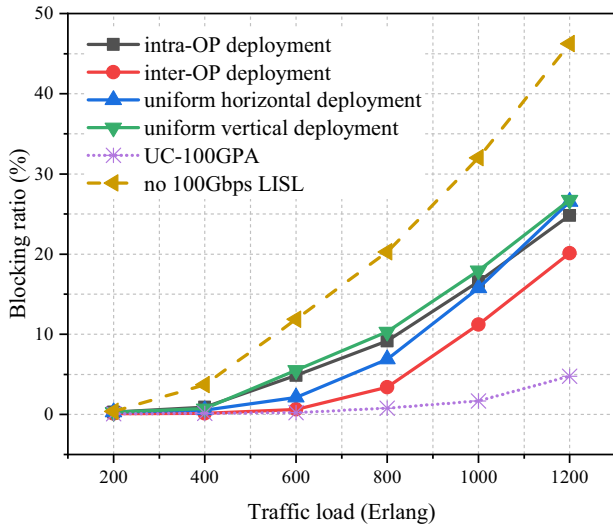


Fig. 4. Blocking ratio under different traffic loads.

100 Gbps LISLs in the intra-OP and inter-OP. In this case, it effectively reduces traffic bottlenecks caused by 10 Gbps LISLs, enhancing the connectivity of 100 Gbps LISLs and fully utilizing transmission advantages of 100 Gbps LISLs. As a result, the UC-100GPA algorithm achieves the lowest blocking ratio.

As shown in Fig. 5, the average latency of all the deployment schemes increases as the traffic load grows. The UC-100GPA algorithm remains the lowest average latency before the traffic load reaches 1000 Erlang. When the traffic load is 1000 Erlang, the average latency of the UC-100GPA algorithm is only 64.66 ms, which is 20.85% lower than the scenario without 100 Gbps LISLs, and 19.19% lower than that of inter-OP deployment. Services do not need to detour frequently and can choose a shorter path under the UC-100GPA algorithm. At a traffic load of 1200 Erlang, the average latency of the UC-100GPA is slightly higher than that of intra-OP deployment. Average latency is computed over successfully transmitted services. Under intra-OP deployment, many requests experiencing high contention are blocked, and the remaining services have lower average latency. In contrast, the UC-100GPA reduces the blocking ratio by admitting more service requests under high traffic load, which results in a slightly higher average latency. This phenomenon reflects a trade-off between the blocking ratio and the average latency under high traffic load.

As shown in Fig. 6, there has been a significant increase in throughput under all deployment schemes compared to the scenario with no 100 Gbps LISL deployment. In this paper, throughput is defined as the ratio of the sum of products of bandwidth and duration of all successful services to the total simulation time. Therefore, throughput has a strong correlation with the blocking ratio. Among the considered deployment schemes, a lower blocking ratio generally leads to a higher throughput. By constructing a fully connected connectivity graph of 100 Gbps LISLs, the UC-100GPA algorithm achieves the highest throughput.

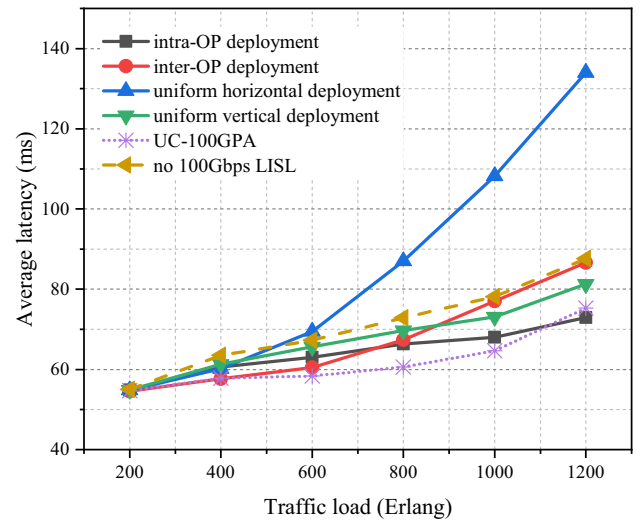


Fig. 5. Average latency under different traffic loads.

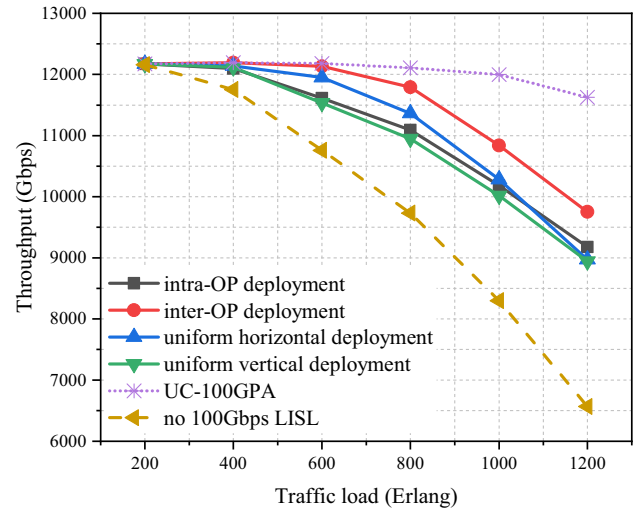


Fig. 6. Throughput under different traffic loads.

C. Results for Different Intra-OP and Inter-OP Ratios of 100 Gbps LISLs

From the analysis in Subsection 5.B, we can see that the network performance of the UC-100GPA algorithm is the best. However, there is a difference in network performance under different intra-OP and inter-OP deployment ratios. This section shows changes in network performance for different intra-OP and inter-OP deployment ratios of 100 Gbps LISLs. To better highlight the differences in the simulation results, traffic load is set to 800 Erlang.

Figure 7 illustrates the blocking ratio under different intra-OP ratios of 100 Gbps LISLs. The comparison includes four scenarios where 100 Gbps LISLs account for 20%, 25%, 30%, and 35% of all LISLs, respectively. As shown in Fig. 7, under different deployment ratios of 100 Gbps LISLs, deploying all 100 Gbps LISLs exclusively in either intra-OP or inter-OP scenarios results in a higher blocking ratio. From a global perspective, the blocking ratio first decreases and then increases as the intra-OP ratio of 100 Gbps LISLs increases. At low

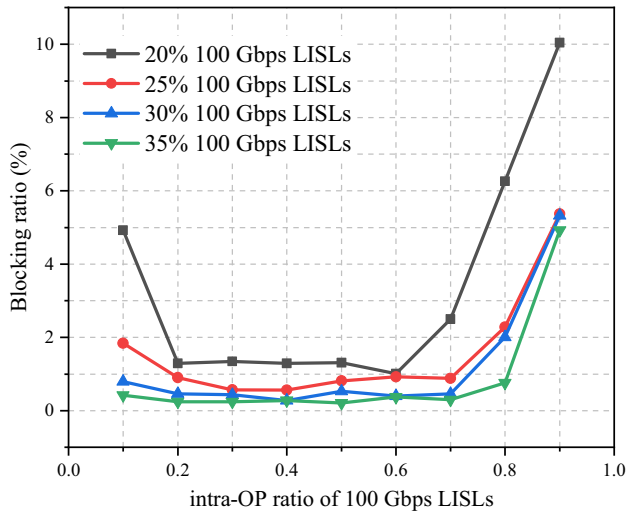


Fig. 7. Blocking ratio under different intra-OP ratios of 100 Gbps LISLs.

intra-OP deployment ratios of 100 Gbps LISLs, increasing the intra-OP ratio enhances local connectivity among satellites, alleviates traffic congestion, and consequently reduces the blocking ratio. However, beyond a certain threshold, excessive concentration of 100 Gbps LISLs within the intra-OP limits the connectivity of the inter-OP, leading to traffic bottlenecks between orbital planes and a subsequent increase in the blocking ratio. In general, the blocking ratio remains relatively low when the intra-OP ratio ranges from 20% to 70%, which indicates that hybrid planning of 100 Gbps LISLs in the intra-OP and inter-OP can significantly reduce the blocking ratio.

Figure 8 illustrates the average latency under different intra-OP ratios of 100 Gbps LISLs. It can be observed that, when the ratio of 100 Gbps LISLs deployed in the intra-OP is between 20% and 70%, the average latency is relatively low. This phenomenon can be attributed to improved network connectivity resulting from a balanced deployment of 100 Gbps LISLs in the intra-OP and inter-OP. Therefore, more service requests can be routed over relatively shorter paths, leading to lower average latency compared with other intra-OP deployment ratios.

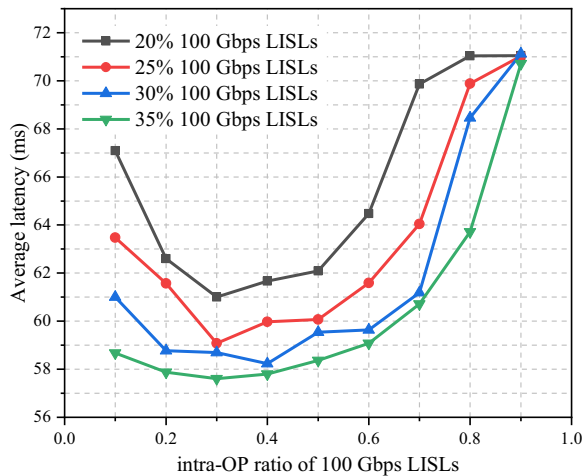


Fig. 8. Average latency under different intra-OP ratios of 100 Gbps LISLs.

From the above analysis, it is evident that the deployment ratio of 100 Gbps LISLs in the intra-OP and inter-OP has a significant impact on network performance. Therefore, the sub-optimal deployment ratio in the intra-OP and inter-OP needs to be determined to ensure that overall network performance is the best. In this paper, the GA-100GPO algorithm is used to verify the changes of the sub-optimal intra-OP deployment ratio and the performance score as the generation increases. The performance score serves as a fitness function for each generation and is defined as the blocking ratio of OSNs. Thus, a smaller performance score implies better network performance. The sub-optimal intra-OP deployment ratio and performance score of each generation are shown in Fig. 9, where the traffic load is 800 Erlang, and the deployment ratio of 100 Gbps LISLs is 25%.

As shown in Fig. 9, the sub-optimal intra-OP deployment ratio changes as the generation increases. When the GA-100GPO algorithm reaches the third generation and beyond, the sub-optimal intra-OP deployment ratio and the performance score remain unchanged, which means that the algorithm reaches convergence and a sub-optimal intra-OP deployment ratio is found. The intra-OP deployment ratio is 63.38%, and the performance score is 0.56, which means that, when 100 Gbps LISLs are deployed according to the ratio of 63.38% in the intra-OP and 36.62% in the inter-OP, network performance is the best. In this case, the blocking ratio is only 0.56%. This is the sub-optimal deployment ratio of 100 Gbps LISLs in the intra-OP and inter-OP.

To examine the impact of traffic load on the GA-100GPO algorithm, we further apply the GA-100GPO algorithm under multiple traffic loads ranging from 1000 to 1800 Erlang, as shown in Fig. 10. With the growing traffic load, the performance score increases accordingly. Meanwhile, results indicate that, although the exact sub-optimal intra-OP deployment ratio varies with different traffic loads, it consistently falls within a relatively narrow range between 50% and 65%.

Notably, the sub-optimal intra-OP ratio obtained at 800 Erlang lies within the same range. This observation suggests that the proposed GA-100GPO algorithm is robust with respect to variations in traffic load, and the conclusions drawn at 800 Erlang remain valid under other moderate-to-heavy traffic conditions.

D. Results for Different Deployment Ratios of 100 Gbps LISLs

In addition to different intra-OP and inter-OP deployment ratios, different deployment ratios of 100 Gbps LISLs can also have a significant impact on network performance. This section shows changes in the network performance for different deployment ratios of 100 Gbps LISLs. Similarly, traffic load in this section is set to 800 Erlang.

As shown in Fig. 11, when the deployment ratio of 100 Gbps LISLs increases, the blocking ratio of different intra-OP deployment ratios shows a downward trend. When the deployment ratio is 50%, the blocking ratio approaches 0, which indicates that increasing the deployment ratio of 100 Gbps LISLs can effectively reduce the blocking ratio.

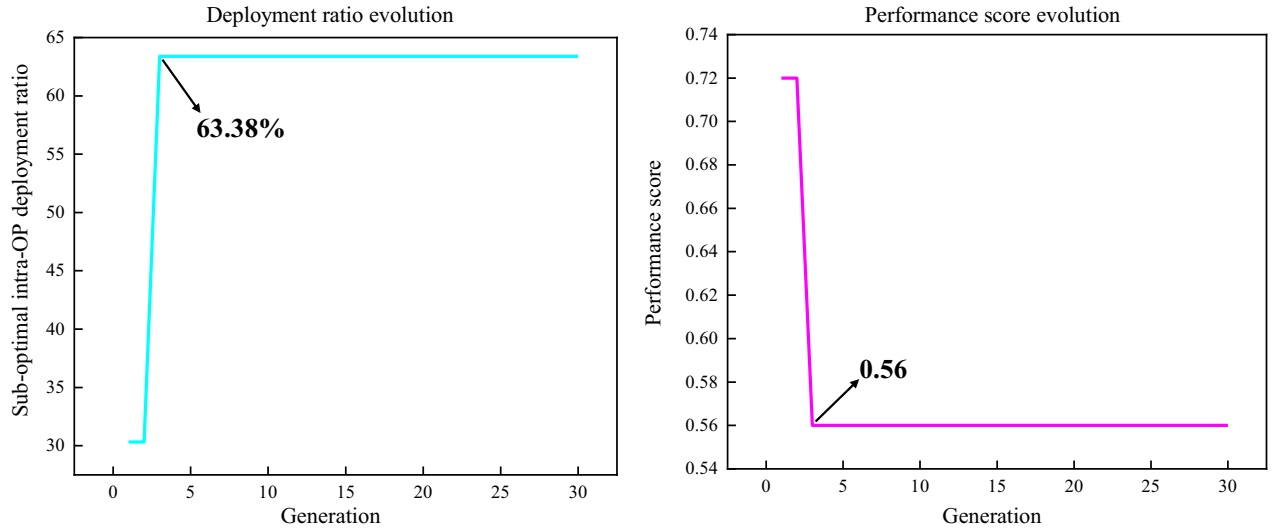


Fig. 9. Changes in sub-optimal intra-OP deployment ratio and performance score.

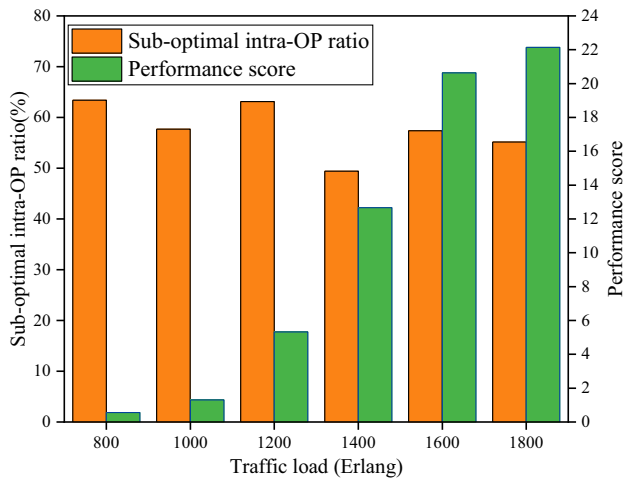


Fig. 10. Sub-optimal intra-OP ratio and performance score under different traffic loads.

However, when all 100 Gbps LISLs are deployed in the intra-OP, the blocking ratio is 8.65%. In this case, although the entire network graph remains connected through 10 Gbps LISLs, the subgraph formed by 100 Gbps LISLs and corresponding satellite nodes cannot form a connected graph when all 100 Gbps LISLs are deployed in the intra-OP. As a result, services that transmit through the inter-OP still rely on 10 Gbps LISLs, making this deployment scheme more susceptible to traffic bottlenecks caused by 10 Gbps LISLs. This situation will inevitably lead to degraded network performance.

Figure 12 shows the average latency under different deployment ratios of 100 Gbps LISLs. When the deployment ratio of 100 Gbps LISLs increases, the average latency for different intra-OP deployment ratios gradually decreases. This phenomenon is attributed to the increased network capacity, which enables most services to be routed along shorter

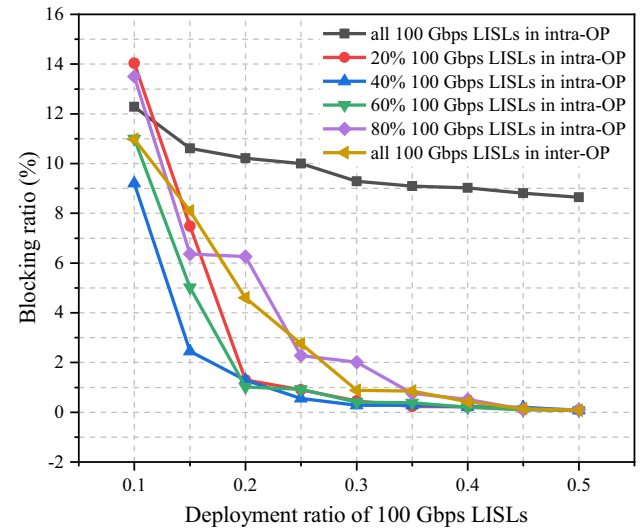


Fig. 11. Blocking ratio under different deployment ratios of 100 Gbps LISLs.

paths, thereby reducing average latency. Consequently, deploying more 100 Gbps LISLs can substantially enhance overall network performance.

E. Discussion on Trade-Off between Cost and Performance

It is evident that increasing the number of 100 Gbps LISLs in OSNs can improve network performance. However, deploying 100 Gbps LISLs inevitably incurs higher deployment costs compared to 10 Gbps LISLs, due to more advanced optical terminals, higher power consumption, and increased system complexity. Therefore, it is important to balance the performance benefits brought by 100 Gbps LISLs against their deployment cost.

As shown in Figs. 11 and 12, increasing the deployment ratio of 100 Gbps LISLs initially results in noticeable improvements in the blocking ratio and average latency. However,

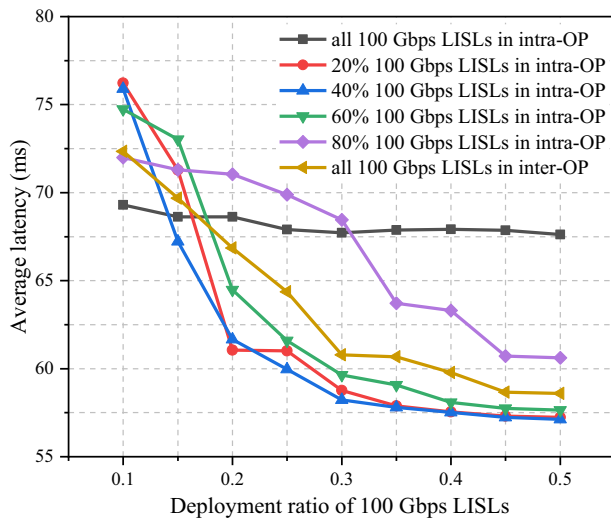


Fig. 12. Average latency under different deployment ratios of 100 Gbps LISLs.

when the deployment ratio of 100 Gbps LISLs exceeds a certain value, the marginal performance gains become limited, and further increasing the number of 100 Gbps LISLs does not lead to significant improvement. In contrast, the deployment cost continues to grow.

These results indicate that there exists a cost–performance trade-off in hybrid 10/100 Gbps LISL planning. An appropriate deployment ratio of 100 Gbps LISLs can achieve most of the performance benefits while avoiding unnecessary deployment cost, which is critical for practical large-scale OSN deployment.

6. CONCLUSION

In this paper, we investigate the network performance degradation caused by the coexistence of 10 and 100 Gbps LISLs in OSNs. To mitigate traffic bottlenecks of 10 Gbps LISLs and improve the utilization of 100 Gbps LISLs, we first propose the UC-100GPA algorithm for the uniform deployment of 100 Gbps LISLs. On this basis, the GA-100GPO algorithm is proposed to further optimize the deployment ratio of 100 Gbps LISLs between the intra-OP and the inter-OP. Simulation results show that the UC-100GPA algorithm significantly reduces the blocking ratio compared to baseline deployment schemes, while GA-100GPO identifies the sub-optimal deployment ratio of 100 Gbps LISLs in both the intra-OP and inter-OP under different traffic loads, further improving the network performance. Moreover, simulation results also reveal a clear trade-off between the blocking ratio and the average latency, as well as the deployment cost and network performance under the scenario of hybrid 10/100 Gbps LISL planning, highlighting the importance of balanced deployment of 100 Gbps LISLs.

However, our work does not consider dynamics or space environmental effects in OSNs. Our future work will take these factors into account, further improving the deployment efficiency and network performance.

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